Henry Lin

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https://github.com/Henry-Linn

Education

University of Wisconsin Madison *Madison, WI*
Master of Science in Physics **Sep 2024 – May 2026**
Relevant Coursework: Quantum Computing, Numerical Linear Algebra, Algebra, Quantum Algorithms, Quantum Mechanics.

University of Massachusetts Amherst *Amherst, MA*
Master of Science in Computer Science **Sep 2023 – May 2024**
Relevant Coursework: Probabilistic Graphical Models, Machine Learning, Reinforcement Learning, Deep Generative Models, Applied Numerical Optimization.

University of Massachusetts Amherst *Amherst, MA*
Bachelor of Science in Computer Science, Physics (Honors), Mathematics **Sep 2020 – May 2023**
Relevant Coursework: Partial Differential Equations, Scientific Computing, Algorithms for Data Science, DataBase, Linear Algebra, Statistic.

Technical Skills

Languages: Python, Java, C++, SQL, MATLAB, JavaScript, Julia, Mathematica, HTML, CSS.
Frameworks: PyTorch, TensorFlow, NumPy, Pandas, Huggingface, Matplotlib, Qiskit, QuTiP, LangChain, Fastapi, Streamlit, Flask, PostgreSQL, SQLite3.
Areas: Machine Learning, Deep Learning, Quantum Computing, Numerical Analysis, Biomedical Imaging, Computer Vision, LLM.
Technology: Git/GitHub, GNU/Linux, AWS Cloud, Docker, Kubernetes.

Experiences

Information Fusion Lab
Undergraduate Researcher *Sep 2021 – Mar 2024*

- Developed unsupervised cardiac MRI segmentation pipelines using the PyTorch deep learning framework.
- Designed custom loss functions for unsupervised learning that incorporated the domain heuristics.
- Conducted experiments, trained deep learning models, analyzed results for the unsupervised cardiac MRI segmentation.
- Achieved a Dice score of 0.70 on the testing dataset, and establishing the first benchmark baseline for future work.

Undergraduate Research Volunteers Program *Dec 2022 – Jan 2023*

- Investigated the latent space of encoded texts and images by using CLIP and deployed the stable diffusion algorithm to generate the image.
- Used linear interpolation to investigate two concepts in the original text embedding space.
- Deployed gradient descent to the distance, and then non-linearly interpolate two concepts in the original text embedding space.
- Implemented cross-attention mechanism to control the stable diffusion algorithm from a research paper.

Papers & Presentations

- **Henry Lin**, Ke Xiao, James Ko, Madalina Fiterau. “Unsupervised Segmentation of Left Ventricle Using Cardiac MRI.” Undergraduate Honors Thesis, 2023.
- Nitya Aryasomayajula, **Henry Lin**, Sneha Pullanoor, Dawn Varughese. “Single/Two-frame(s) Volume Estimation of the Left Ventricle Using Convolutional Neural Networks.”
- **Henry Lin**, Pracha Promthaw, Nikhil Gautam, Dmitry Petrov. “Continuous Control of Latent Diffusion Model Through Token Embedding Interpolation.”

Awards

- **YQuantum hackathon:** 2025, Yale University, Yale Hackathon BlueQubit challenge winner.
- **The Blaise Pascal Quantum Challenge:** 2025, Pascal, top 15 teams invited to Pascal Quantum challenge under the team name: quaNtumFix. Project is about Ammonia production with quantum simulations for green agriculture.
- **Bay State Fellowship:** 2023-2024, University of Massachusetts Amherst, competitive fellowship awarding a teaching assistantship position, providing full tuition waiver, health insurance waiver, and stipend for up to four semesters for a master’s degree in computer science.
- **High Demand Scholarship:** 2022-2023, University of Massachusetts Amherst, awarded for academic achievement for STEM undergraduate students.